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PHYSICS
Paper 1
(Theory)
July/Aug. 2026
2 Hours

AITEL JOINT MOCKS ASSESSMENTS 2026

Uganda Certificate of Education

PHYSICS

Paper 1

(Theory)

2 Hours 30 Minutes

INSTRUCTIONS TO CANDIDATES:

*This examination paper consists of two Sections; **A and B.***

It has seven examination items.

Section A has three compulsory items.

*Section B has **two Parts; I and II.** Respond to **only one** item from **each part.***

*Respond to **five** items in all.*

*Any additional item(s) responded to will **not be scored.***

***All responses** must be written in the answer booklet(s) provided.*

*Responses to each item should start on a **fresh page.***

*Indicate the item number you are responding to, **clearly at the top of the page.***

Graph paper is provided.

Mathematical tables and silent non-programmable calculators may be used.

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SECTION A

*Respond to **all** the items in this section.*

Item 1:

During a cultural festival held at night, a musician performed on a floating raft in the middle of a wide river. Around the stage, powerful **laser beams in red, blue, and green** swept across the water surface, creating dazzling reflections. The audience on the banks and boats wore bright yellow outfits decorated with black patterns, and they were astonished when the flashing lights seemed to change the appearance of their clothes.

The guitar produced sound waves traveling through air at a frequency of **440 Hz**, while organizers projected strong beams of light across the river to improve visibility. The riverbank was about **1 km away** from the raft. Two boys, A and B, standing along the same line of sight to the stage, noticed something unusual: Boy A heard the sound after **3 seconds**, while Boy B heard it after **5 seconds**, which made them curious to investigate the event.

Hint: Speed of sound in air = 330 m s^{-1}. Speed of light in air = 3.0×10^{8} m s^{-1}.

Task

(a) Explain why the boys heard sound differently at different intervals of time.

(b) Describe why the clothes appeared to change colour when illuminated by flashing lights.

(c) Justify why laser beams were preferred to enhance visibility at night along the riverbank.

(d) Compare the wavelength of sound and laser light in the air medium.

Item 2

An international logistics hub utilizes an advanced electromagnetic scanner to inspect incoming cargo. During a routine shift, operators need to screen two distinct items: a thick, heavy-duty metallic shipping container suspected of containing dense contraband, and a lightweight piece of passenger luggage filled with delicate electronic gadgets.

The facility's scanner operates at a fixed frequency of 6×10^{18} Hz. The technical manual provides the following reference data for the scanner's imaging software:

System Calibration Reference:

Wavelength	Image Resolution

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5×10^{-11} m	High
5×10^{-10} m	Medium
5×10^{-9} m	Low

Task:

As a physics consultant called in to optimize the scanning operations, address the following:

(a) Determine the operational wavelength of the scanner and comment on the expected clarity of the generated scan images for the cargo.

(b) (i) Explain how the frequency and wavelength of these electromagnetic waves determine their ability to pass through dense materials.

(ii) How does this physics principle dictate how the scanner should be configured differently for the heavy metallic container versus the delicate passenger luggage?

(c) Outline the critical occupational safety measures and protocols that operators must adhere to while working around high-energy scanning equipment.

Item 3

During a weekend afternoon at around 2:45 pm while relaxing in their living room, a family watched a live international football match being played amidst a heavy snowstorm in a distant continent on a satellite sports channel. Interestingly, the match was taking place deep into the night under stadium floodlights, and the players were visibly struggling with freezing temperatures. The family members wondered how it could be pitch dark and snowing heavily in that foreign stadium, yet it was a hot afternoon with the Sun shining brilliantly outside their own house.

Task:

As a physics learner, help the family clear their queries about:

a) The occurrence of a freezing snowstorm in that distant country while it is a hot, clear afternoon at their home at the very same time.

b) Why it was night-time in that foreign country yet it was broad daylight in their local area.

c) How the live television signals broadcast from the stadium across the world managed to reach their TV set at home.

SECTION B

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Part I

*Respond to **one** item from this part.*

Item 4

A local entrepreneur is constructing a three-story commercial complex in a fast-growing urban trading center. Currently, the site workers manually pull up heavy buckets of wet concrete mix to the top floor, which is at a height of 12 meters, using a thick sisal rope.

During a site visit, the entrepreneur finds the workers complaining bitterly. They explain that pulling the rope repeatedly makes it scorching hot, and over weeks of work, their palms have become extremely rough, hard, and blistered. Additionally, they get deeply exhausted after lifting only a few buckets and have to take frequent breaks, slowing down construction. The site foreman advises the entrepreneur to procure a motorized hoist or a mechanical pulley system to lift the concrete buckets efficiently, but the entrepreneur does not understand how such a machine functions or how it would solve the workers' problems.

Hint: *1 litre of concrete mix=1000 cm^3, Density of wet concrete=2400 kg m^{-3}, Acceleration due to gravity, g=10 ms^{-2}*

Task:

As a learner of Physics;

a) Explain to the construction workers why the rope becomes hot during continuous use and why their hands have become hard and rough.
b) Explain to the workers the physics behind why they get easily exhausted when lifting the heavy concrete buckets to the top floor.
c) Help the entrepreneur understand the type of machine needed for this task and clearly explain how it works to solve the site's challenges.

Item 5

An apprentice mechanic at a service garage needs to clean a heavily greased engine block using a high-pressure hot-water spray wash. To effectively melt the grease, the mechanic needs to prepare 200 cm^3 of water at a temperature of 80°C. To reach this temperature, the water must be heated inside a compact chamber using a built-in 1200 W electrical heating element. The garage owner states that for the cleaning chemical mixture to remain stable, this heating phase must be completed within 40 s.

Furthermore, the manual spray gun relies on a small hand-pumped piston mechanism to pressurize and eject the hot water. The apprentice is unfamiliar with how this pump assembly operates to draw and discharge fluids.

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Hint: *Specific Heat Capacity of water=4200 J kg^{-1} K^{-1}*
Initial Temperature of water=25°C

Density of water=1 g cm^{-3}

The spray gun pump has a piston cross-sectional area (A)=3.0 cm^2

The hot water can only blast through the spring-loaded nozzle valve if the generated pressure inside the cylinder is greater than 45 kPa.

Task:

As a learner of Physics;

(a) Help the apprentice mechanic if the water will reach the required temperature within the owner's specified time limit.

(b) Explain to the apprentice how the piston pump mechanism works to draw water from the reservoir and force it out through the nozzle.

(c) Establish if an effort force of 15 N applied by the mechanic on the pump's piston handle would allow the hot water to blast through the nozzle valve.

Part II

*Respond to **one** item from this part.*

Item 6

The owner of a mobile food truck connects the vehicle to a 240 V AC mains supply at a central food park. Inside the truck, the owner intends to install several appliances designed for a 120 V internal system configuration: a commercial deep fryer rated 120 V, 3000 W, a microwave oven rated 120 V, 1200 W, an electromagnetic order-ready buzzer with a resistance of 8 Ω that instantly sounds to alert staff when a counter button is pressed, and a string of 6 decorative LED spotlights each rated 120 V,15 W for night-time illumination.

To power the cooking counter, the owner buys a heavy-duty extension power strip fitted with a safety fuse rated at 13 A, intending to plug both the commercial deep fryer and the microwave oven into this single extension strip.

Task:

As a learner of Physics;

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(a) Comment on the effectiveness of the 13 A fuse in the extension power strip if the owner runs both the deep fryer and the microwave oven altogether at the same time.
(b) Advise the truck owner on the best configuration to connect the 6 LED spotlights around the vehicle and give a reason for your choice.
(c) Explain the step-by-step physical process of how sound is produced by the electromagnetic buzzer system when the counter button switch is pressed.
(d) Determine the total financial cost required to keep only the microwave oven operating for a cumulative total of 3 hours per day, given that the electricity tariff rate at the food park is UGX 750 per kWh.

Item 7

A community radio station situated on an elevated hill was recently affected by a severe thunderstorm. The electrical surge from a nearby lightning strike caused widespread damage inside the transmission room, including a blown main circuit fuse, a melted copper grounding strap, and a malfunctioning power transformer that served as the station's primary voltage regulator.

The original power regulation system was required to deliver a steady power output of 100 W to keep the transmitter broadcasting at full range. Short on funds, the station manager decides to replace only the damaged transformer so they can resume broadcasting immediately. However, the manager has refused to replace the damaged lightning rod on the transmission tower or to re-establish any proper earthing connections to the ground.

Hint: *The replacement transformer is designed to work on a 240 V AC supply with an input power rating of 120 W.*

Its primary coil has 600 turns, its secondary coil has 400 turns, and it operates at an efficiency of 75%.

Task:

As a student of physics, explain to the station manager;

(a) (i) How installing a functional lightning conductor on the radio tower will protect the station's infrastructure during future storms.
(ii) Why a proper earthing connection is essential for both the structural safety of the equipment and the overall performance of the transmission system.
(b) Determine whether the newly acquired transformer is suitable for meeting the station's broadcasting power regulation needs. (Support your answer with relevant calculations).

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(c) Discuss the critical hazards and dangers associated with running the radio station's electrical equipment without replacing the blown fuse and failing to establish a secure earthing network.

END

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